

1903

Largest Odd Number in String

Dry Run

num = "3542786"

logic $\rightarrow$ if number ends with
1/3/5/7/9 so it
is odd number.

(I)

3 5 4 2 7 8 6
 ↑ ↑
 i i
 else else

check $i == 1/3/5/7/9$
if yes $\Rightarrow$ return

else

check ~~to~~ the ~~to~~ upcoming left
no.

at $i \rightarrow 3$ {idx}

$i == 7$

So we need to return
for the string starting
at this idx.

To get this substring
we use inbuilt substring
method

`num.substring(0, i+1);`

here we get the
largest odd number in
the string

code $\Rightarrow$ T.C $\Rightarrow O(n)$ S.C $\Rightarrow O(1)$

String helper(String num) {

 for(int i = num.length()-1; i >= 0; i--) {

~~11/11~~

char ~~at~~ digit = num.charAt(i);

if (digit == 1 || digit == 3 ||
digit == 5 || digit == 7 ||
digit == 9) {

return num.substring(0, i+1);

}
return " ";

if there are
no odd numbers
in the string